CANDIDATE NAME (CG) ________________________ INDEX NUMBER _________

SERANGOON JUNIOR COLLEGE
PRELIMINARY EXAMINATION 2008

BIOLOGY
Higher 1

Paper 2- 8875

Saturday
23 August 2008

2 hours

Additional materials:
Answer paper

READ THESE INSTRUCTIONS FIRST

Write your name and index number in the spaces at the top of this page and on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Write your answers in spaces provided on the question paper.

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

All working for numerical answers must be shown.

At the end of examination,

1. fasten all your work securely together;
2. enter the numbers of the Section B and C questions you have answered in the grid opposite.

FOR EXAMINER'S USE	
Section A	**/30**
Section B	
1	**/10**
2	**/10**
3	**/10**
4	**/10**
Section C	
	/20
TOTAL	**/90**

INFORMATION FOR CANDIDATES

The intended number of marks is given in brackets [ ] at the end of each question or part question.

This question paper consists of 9 printed pages.

Section A (40 marks)

Answer all questions in the space provided

Question 1

When buds of the morning glory flower, *Ipomoea*, open into flowers, the pH in the vacuoles of the flower epidermal cells increases. The functional ion pump that brings about the change in pH of the vacuole is coded by a **gene A**. However, a mutant morning glory that has a genotype of **aa** codes for a non functional membrane – bound ion pump.

Normal blue flowers have a coloured pigment anthocyanin, coded by the dominant allele of the **gene B**. Plants with the genotype **bb** cannot produce anthocyanin and they have white flowers.

(a) List the genotypes of morning glory plants that result in

(i) Blue flowers with functional ion pumps[1]

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(ii) White flowers with non-functional ion pumps [1]

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(b) Morning glory plants with the genotype AABB and aabb were crossed. Draw a genetic diagram to explain the cross and give the ratio of phenotypes expected in the F1 generation [4]

The mutant allele of the gene A has a single base deletion which produces a non functional ion pump that is unable to increase the pH of the vacuoles.

(c) Describe how a single base deletion in the mutant allele leads to structural and functional changes of the inactive ion pump. [4]

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[Total: 10 marks]

Question 2

2,6-dichlorophenolindophenol (DCPIP) is a blue dye that can be converted to colourless reduced DCPIP by gaining electrons. This is summarized below.

$$\text{DCPIP (blue)} \xrightarrow[\textit{electrons}]{} \text{Reduced DCPIP (colourless)}$$

A chloroplast suspension was made by grinding fresh leaves in buffer solution and centrifuging the mixture. Tubes were prepared and treated in different ways. The colour of the tube contents was recorded at the start and after 15 minutes. This information is summarized in the table.

			Colour	
Tube	**Contents**	**Treatment**	**At start**	**After 15 minutes**
A	$2cm^3$ chloroplast suspension $6cm^3$ DCPIP	Tube kept in bright light	Blue/ green	Green
B	$2cm^3$ chloroplast suspension $6cm^3$ DCPIP	Tube kept in dark	Blue/ green	Blue/ green
C	$2cm^3$ buffer solution $6cm^3$ DCPIP	Tube kept in bright light	Blue	Blue

(a) (i) Tube C was included as a control. Explain why this control was necessary in the investigation. [1]

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(ii) Explain the colour of tube A after 15 minutes. [2]

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(b) Describe how reduced NADP is involved in the light-independent reactions of photosynthesis. [2]

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(c) Apart from the difference in the presence or absence of oxygen, list a difference between aerobic and anaerobic respiration. [1]

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(d) During glycolysis, NAD is reduced. Explain what happens to this reduces NAD when the cell is respiring anaerobically. [2]

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[Turn over

(e) Explain why, in the absence of oxygen, the electron transfer chain in the mitochondria does not operate. [2]

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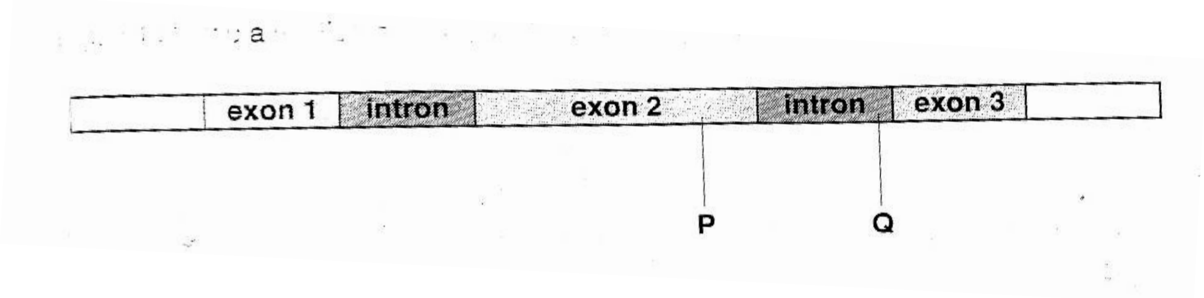
[Total: 10 marks]

Question 3

Figure 3.1 shows a section of DNA, containing one gene.



(a) (i) State one way in which the structure of this gene indicates that it is from a eukaryote. [1]

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(ii) Using the same diagrammatic format as Fig 3.1, show what the pre-mRNA and mature mRNA will be for this gene.

Pre-mRNA [1]

Mature mRNA [1]

(iii) Describe the process that has occurred to when a pre-mRNA is converted to a mature mRNA. [3]

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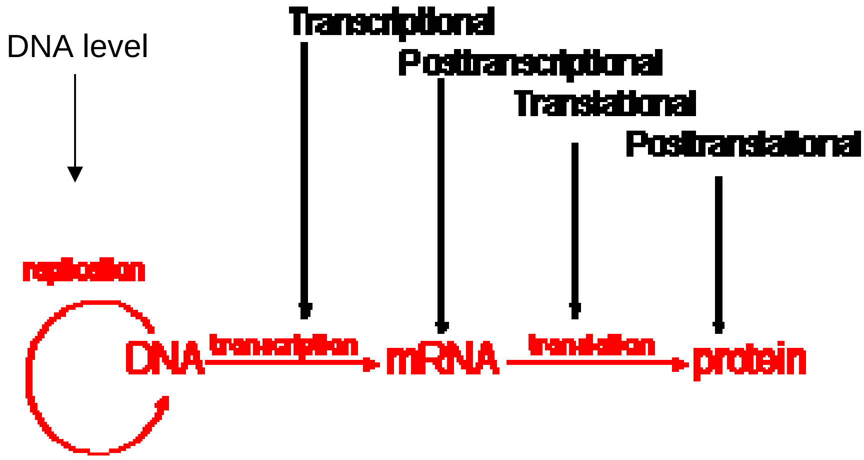
Control of gene expression can occur at any step: at the DNA level, transcriptional level, post-transcriptional, translational level and post-translational level.



(iv) State two ways to control the expression of genes at the DNA level and explain how these modifications work to control gene expression. [4]

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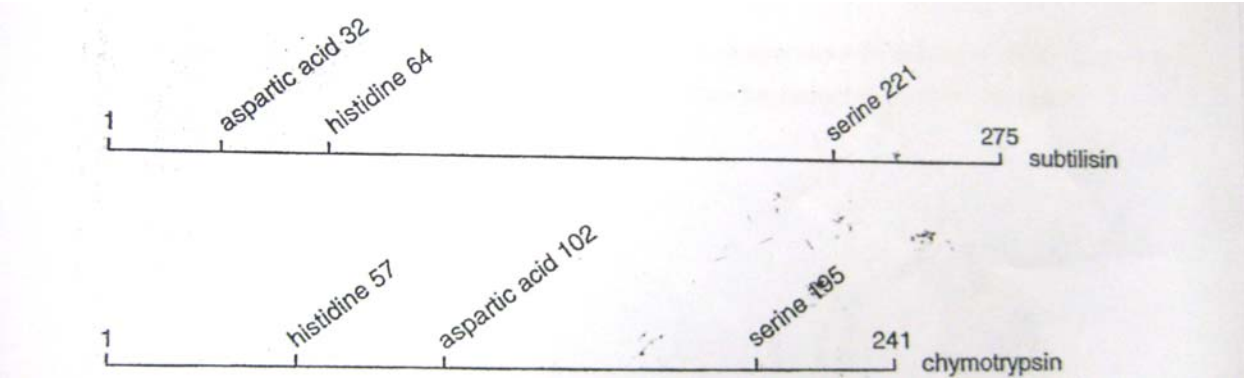
[Turn over

Question 4

The enzyme subtilisin is a protease synthesized by bacteria and is made of 275 amino acids. The enzyme chymotrypsin is a protease synthesized by mammals and is made of 241 amino acids. Both enzymes have the same arrangement of 3 amino acids, serine, histidine and aspartic acids in their active sites but they are structurally different with the three amino acids being in different positions in the amino acid sequence as shown in Fig 4.1.



(a) Describe how amino acid residues at different positions in the protein may be brought together in the active site when the enzymes are synthesized. [4]

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(b) Explain why having the same arrangement of amino acids in their active sites allows these two different enzymes to catalyze the same reaction. [3]

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Imagine that, on a planet in a distant galaxy, evolution has resulted in organisms which the same basic biochemistry as humans. However, the organisms have 6 rather than 4 different bases of their DNA, with a base composition: 24.5% D, 19% E, 25.2% F, 7.4% G, 7% Z and 18.9% X. They have proteins built out of 30 rather than 20 amino acids.

(c) With reference to the information given above:

(i) Deduce the DNA structure of these extra-galactic organisms. [2]

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(ii) State, with a reason, the minimum number of nucleotides that code for one amino acid. [1]

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[Total: 10m]

Section B (20 marks)

Answer any **only one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be set out in **sections (a), (b)** etc., as indicated in the question.

<u>Question 5</u>

(a) Outline the procedure and discuss the underlying principle for the process of Southern blotting. [15]

(b) Compare the process of Western and Southern blotting. [5]

<u>Question 6</u>

(a) Compare and contrast between adult stem cells and embryonic stem cells [6]

(b) State what is meant by the term genetic engineering and explain how genetic engineering can improve the quality and yield of crop plants and animals. [14]